



Computers & composition research at the dawn of generative AI: Threats, opportunities & future directions

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ABSTRACT

Generative AI systems hold potential as assistive technologies throughout the research process, but their usage also surfaces a number of issues researchers need to consider. In this article, I begin with the threats to research and researchers raised by genAI (and threat is not too strong of a word). I then address the opportunities for AI assistance throughout the research process from study planning and data collection to data analysis and reporting. I close by raising some future research directions for our field to consider, so that we have a voice in shaping the design and usage of AI for writing research.

In the inaugural issue of *Computers and Composition*, editors Cynthia Selfe and Kate Kiefer (1983) provided a list of research questions for readers to consider in relation to a new technology, *computers*, including their last question: “What steps are businesses and the computer industry taking that will affect our teaching of writing in the next ten to twenty years? Will research in artificial intelligence change the way we think about communicating with each other and with computers?” (p. 1). Two pages later, Hugh Burns (1983) reflected on his ground-breaking research in artificial intelligence that was also the subject of his dissertation (Burns, 1979) and his work related to the writer invention program TOPOI, explaining how natural language processing worked at the time and speculating on areas for further development. But, as he noted, “each improvement [in AI processing] has both good and bad consequences the humanistic composition teacher should consider” (p. 3).

Here, on the occasion of the 40th-anniversary issue, I take up these prescient questions and reflections from the first issue with a focus on research instead of pedagogy. How is computers and composition research and how are we as researchers going to be and already are being impacted by AI? What good and bad consequences, to use Hugh Burns’s words, await us as genAI continues to develop, and its usage becomes even more prevalent? Just as in 1983 when Cindy Selfe and Kate Kiefer and so many others were on the cusp of a new age of writing—the age of word processing and then networked computing—so too are we now entering a new age, the age of AI. But, then again, maybe it’s not all new. And there’s the crux of the matter. Even as we feel like we’re drinking from a firehose of AI Hype (and AI Armageddon), we have much we can draw from in our field’s approaches to researching new writing technologies that can help guide us going forward.

From its outset in the pages of *Computers and Composition* and in its constellation of related publications, our field has recognized that computerized technologies have uses and impacts that we need to study so as to develop the knowledge and evidence-based findings that can help shape future integrations of technology for writing, the teaching and learning of writing, and the research of writing. With evolving theoretical frames and methodological approaches, scholars in computers and writing consistently recognize that technologies occur in (and pick your frame) activity systems, ecologies, networks, relationships, embodied materialities, and more and that to study any technology for writing necessitates considering the contexts shaping usage. We also as a field have, with varying

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degrees of focus, emphasized the ethics of technology design and usage and the ethics of studying with and about digital technologies. We have applied, nearly from the outset, critical frames for understanding technology, reflecting on the biases, power dynamics and impacts of corporate entities, government surveillance, biased designs and, later, algorithmic outputs. The field of computers and composition, as well as the journal itself, has tried, not always successfully, to ensure that the experiences and perspectives of people from traditionally marginalized communities are central to the field as scholars and as participants for whom research may bring benefits and change. And our field has consistently explored what it means to research with digital technologies and how emerging technologies create new opportunities and new challenges to consider. In relation to that latter point, publications such as *Opening Spaces: Writing Technologies and Critical Research Practices* (Sullivan & Porter, 1997) and the collection *Digital Writing Research: Technologies, Methodologies, and Ethical Issues* (DeVoss & McKee, 2007) provide early critical frames and in-depth perspectives on research in the field and how researchers approach their work researching with and about digital technologies, focusing on such things as word-processing, networked computing, online discussion forums, IRBs with digital research studies, time-use diaries and video screen capture to study writing processes and systems, considering global networks for writing and more.

And now we face a new and emerging technology to consider—*artificial intelligence*. In this article, I focus on large-language model generative AI. I begin with the threats raised by genAI (and threat is not too strong of a word), moving to consider the opportunities for AI assistance throughout the research process (from study planning and data collection to data analysis and reporting) and raising some research questions our field might consider now and in the future so that we have a voice in shaping the design and usage of genAI technology. There are a lot of corporations designing, building, and hyping AI writing systems without very much knowledge of rhetoric, writing, or composing processes and pedagogy. Our research has a lot to offer—and, further, we have a responsibility to provide it. Just as the founders of the field created a space for research on technologies of writing, so too must we take up the challenge of genAI by researching it, identifying its flaws, harms, and problems; locating its strengths and advantages; and influencing its future design and usage in research. This article takes up that challenge by considering how we can approach AI in research now and in the future.

Threats to research raised by genAI usage

The threats posed by genAI usage to research and researchers are real and growing. As researchers we individually and collectively need to be aware of these threats and seek ways to address them or else the future for our research may not be sustainable.

Dangers of faked research

I begin with a serious issue that perhaps has not hit our research field as much—yet. But it will, and it's not hyperbolic to say that the very future of all research is at stake. AI is being used in malevolent and dishonest ways to produce fake research—and at a scale that is flooding publication venues.

If you haven't searched your publications in common venues like Google Scholar of late, you may wish to do so. It could be you might find, as did Diomidis Spinellis (2025), that you've authored an article you had no idea you'd researched or written because, well, you hadn't. When Spinellis asked the *Global International Journal of Innovative Research (GIJIR)* to take down the fake published article that he did not author, the journal did so, but the DOI was (at the time of publication) still registered and the Internet Archive's Wayback Machine had an archived copy. In looking at more articles in *GIJIR*, Spinellis found at least 48 that appeared fake, and, as he explained, misattributed and fake articles "threaten the credibility of academic publishing" (np). But it's not just credibility under threat. The foundations upon which fields build their knowledge and make recommendations, advise on policy etc. are being shaken and manipulated through AI usage. Dip your toe into the world of falsified research study scholarship and you realize quickly that *Here be dragons* for sure.

A study published in the Harvard Kennedy School *Misinformation Review* found that approximately two-thirds of papers retrieved from common data bases such as Google Scholar were "produced, at least in part, through undisclosed, potentially deceptive use of GPT" and that the "majority (57 %) of those questionable papers dealt with policy-relevant subjects (i.e., environment, health, computing) susceptible to influence operations" (Haider et al., 2024). They also found that most of the publications were available across different domains such as social media, archives, and repositories, so as to spread the misinformation widely. Their findings are similar to many other studies looking into faked research, including one study that examined "copy-cat" papers, where AI was used to rework an already published paper or AI was used to crank out redundant publications from publicly available data. Maupin et al. (2025), for example, found over 400 papers on biomedicine generated by AI in 112 journals. An article in *Nature* about this study (Naddaf, 2025a) quoted a researcher who explained, "If left unaddressed, this AI-based approach can be applied to all sorts of open-access databases, generating far more papers than anyone can imagine [...] This could open up Pandora's box [and] the literature may be flooded with synthetic papers" (np).

Slugging through digital reams of faked papers wastes researchers' time, for sure, but it also poses broader societal risks. As Haider et al. (2024) noted:

Two main risks arise from the increasingly common use of GPT to (mass-)produce fake, scientific publications. First, the abundance of fabricated "studies" seeping into all areas of the research infrastructure threatens to overwhelm the scholarly communication system and jeopardize the integrity of the scientific record. A second risk lies in the increased possibility that convincingly scientific-looking content was in fact deceitfully created with AI tools and is also optimized to be retrieved by

publicly available academic search engines, particularly Google Scholar. However small, this possibility and awareness of it risks undermining the basis for trust in scientific knowledge and poses serious societal risks.

Haider et al. (2024) and Maupin et al. (2025) were focusing in particular on health-related research. For us as writing researchers, our work may not be at the top of the mind for malicious influence operators. In the United States we hear a lot more about the CDC under Robert Kennedy Jr. producing studies with fake citations (Dall, 2025; Prada, 2025). In fact, medical researchers were some of the first to raise the alarm about AI being used to fake studies on health (Májovský et al., 2023). But for writing, we need to remember that there is money to be made in creating and selling AI platforms and systems for writing to institutions. Higher education institutions might be quite eager to replace the expensive-to-staff many sections of courses such as first-year writing, technical writing, and business writing with the newest, latest AI platform backed by “research” showing how these 24/7 AI tutoring platforms are more effective and efficient than human writing teachers (see Sano-Franchini et al., 2025 for discussion of AI hype masquerading as research). We as a field have always had to counterbalance industry claims with our own in-the-field research (e.g., Ericsson & Haswell, 2006). Now, in addition to considering fake scholarship produced with a corporate bias, we need also to be alert for fake research produced by malicious actors seeking to undermine more rigorous and ethically responsible research.

To counter the problems of faked research, Spinellis (2025) raised the need for “strengthening identity verification mechanisms for DOIs and ORCIDs, enhancing AI detection methods, and reforming research assessment practices” (np). I am leery of AI detection methods since so many of those so-called “detection systems” misidentify human-written text and, in the classroom particularly, create a pedagogy of distrust rather than trust, stifling transparent and open dialogues about how to use and work with AI and how and when not to (for more on surveillance technologies and AI detection systems, see Fernandes & McIntyre, 2025; McIntyre, 2025). But at the scale that AI is flooding the publication world, I can see how there needs to be more up-front gatekeeping, but such gatekeeping comes at a cost.

Harms to the research publication & review process

Given the flood of synthetic papers, produced by AI working with real data or faked data, journals and other publication venues are getting overwhelmed with some having to change their policies. For example, the free open-access platform *arXiv* that has published over 2 million articles in science, math, and computer science has had to change policies for the computer science (CS) category and specifically for review articles and position papers because they now receive hundreds of review articles each month (Boboris, 2025). Before being considered for submission in the CS category, review articles and position papers must now show proof of having been accepted at a journal or a conference and completed successful peer review. *arXiv* made this change because, as they explained:

The advent of large language models have made this type of content relatively easy to churn out on demand, and [...] our team of volunteer moderators do not have the time or bandwidth to review the hundreds of these articles we receive [...] Reasonable and trusted outside refereed venues already exist (conferences and journals) which solicit position papers and review articles on subjects of concern or interest to our readers [...] and as part of that process, they conduct in-depth review to assure quality, evidential support of opinions, and scholarly value. (Boboris, 2025)

The *arXiv* team is putting their trust in the fact that human reviewers will be able to identify and reject for publication AI written research or research with faked AI-generated data. But that may not be the case, as blind deception studies have shown where human reviewers were not able to reliably distinguish between human-authored and AI-generated manuscripts (Helgeson et al., 2025).

But increasingly editors, in their initial determination of whether a manuscript should be sent out to review, and reviewers in their role as readers and assessors of a manuscript’s quality, are having to grapple with questions such as is the data presented even real? Did this study actually happen? Is the article even written by a human? This raises a key question: can the human peer-review process work as a check against fake studies? One problem is that the peer review process was never designed as a check to see if the data were actually *real*. There was always a good deal of trust involved at that level. What held the system together was that if falsification of evidence were discovered, a researcher’s credibility (and, likely, employment) would be irreparably damaged. But like the flood of misinformation online, at some point if the torrent of AI-generated material is too strong, how will the non-AI generated manuscripts find the time and space to rise to publication?

Being inundated with a flood of manuscripts would be challenging for any publication venue, even one such as *Computers and Composition* with a deep pool of reviewers. Reviewers have a lot to consider and a careful review that aims to be supportive and provide substantive and engaged feedback to the authors takes a lot of time. But as we face changing institutional governance and changing faculty workload policies, where faculty are expected to do more teaching, produce more research, engage in more campus service, finding the time and the career reward for doing reviews is getting more challenging, especially if universities are not valuing or crediting external service work. Some see genAI itself as the answer to this crisis—let’s use AI to conduct reviews (and AI is beginning to be used to review grant applications, see Adams, 2025). But the same criticism about genAI’s use in creating research articles applies to its use in peer review: the genAI system does not have experiential knowledge and cannot provide the critical perspective of an engaged human reader with deep expertise and contextual understanding of a field’s evolving theories, methods, and research questions and nuanced consideration of research populations and ethics of engagement. In addition, using genAI for peer reviews and uploading a researcher’s manuscript to an AI system without their consent is problematic and unethical. As I will discuss more below, the harms to intellectual property and the dangers to research confidentiality are significant and must always be considered.

Haider et al. (2024) don’t advocate for AI for peer reviews, but they do mention sites and platforms that identify AI-related articles, often through AI detection systems. Academ-ai.info provides tell-tale ChatGPT phrases quoted from some flagged articles, such as: “As

an artificial intelligence, I cannot provide”; “As of my last update”; “Certainly, here are...” But reading the entries on some of these AI-flagging sites, it’s not clear if an article listed was written entirely by genAI or if a researcher simply overrelied on it for some sections and, in the case of the phrases quoted, did a bad job copy editing their prose. As a field focused on computers and composition, we have a lot to contribute to these discussions about what use of genAI might be appropriate in research writing and how AI usage might be assessed and viewed in the research review process. But we also need to be attuned as editors and reviewers, to the types of submissions we are seeing in our publication venues, including to *Computers and Composition* and the constellation of related publications such as *Computers and Composition Digital Press*. If seemingly AI-generated submissions begin to come in at rates that challenge or overwhelm the system, we will need to consider approaches to adapt.

I want to make one last point related to faked research that we will need to be aware of now and in the future: the threat to our own research findings and literature reviews. With so much more faked research out there, as we conduct research, particularly in relation to computers and writing where often we need to gather from interdisciplinary perspectives on technology, we might find ourselves using and referring to faked research without realizing it. This happened to me recently when I was drafting AI research guidance for Miami University’s Office of Research & Innovation. I was drafting a section on the errors in large language models (LLMs) and the problems of LLM design, and I cited an article from *arXiv*, a pre-print that had not yet been accepted. At this point I was not aware of the flood of fake articles overwhelming *arXiv*. Luckily, when sharing the draft of the write-up with colleagues, a colleague in Computer Science alerted me that source was not considered reliable and people were no longer citing it. Now I recognize that I will need to do a deeper verification process when researching to ensure that the articles or pre-prints I’m looking at have at least undergone peer review and are being published in a reputable venue. But, again, this relies on the human-based peer review process to catch faked scholarship and, well, that may not always be possible. So, it behooves all of us going forward in our research to be extra careful when using platforms like Google Scholar or *arXiv* that have a more open inclusion policy. The bottom line is we need to be aware of and keep track of the rise of faked research and prepare ourselves and our publication venue review processes for how to address these threats.

Challenges in the design of LLMs

So far, I’ve been focused on threats to research publication and to the research review process, but now I want to turn briefly to challenges that arise by the very design of the large-language models (LLMs) that form the basis for most genAI systems.

Errors

Anyone who chooses to work with generative AI, must remember that there is no human intelligence and certainly no ethical reasoning in the massive data sets and algorithmic analyses and outputs produced by generative AI systems (whether text or image or audio). LLMs generate their outputs by predictive analytics which makes the systems vulnerable to errors arising first, of course, from their original training sets and their initial programming (Bender et al., 2021). But as they get larger with more parameters the errors don’t actually lessen; in fact, the larger they get the more LLMs make things up and make erroneous connections; it’s an innate flaw in the design of LLMs and having more data and larger models doesn’t fix it (Kalai et al., 2025; Xu et al., 2025). In a fascinating study conducted by the BBC, they found that even summarizing news articles, which are far less complicated and detailed than research articles, many genAI platforms introduced errors (Elliott, 2025), including making up quotations or misrepresenting viewpoints. Error in outputs are inevitable, and researchers choosing to work with AI have to plan for and address that through human-centered verification processes.

Biases

Because of biases in their training data, LLMs can produce biased outputs in relation to text (Hofmann et al., 2024) and other multimodal data (Kotwal & Marcel, 2025). These biases can arise at any stage of the research process, in proposing study design, in planning data collection, in analyzing data, and in writing up data. Researchers from across fields have found that genAI, particularly off-the-shelf LLMs that have not been customized (e.g., ChatGPT, Google Gemini), introduce extensive biases into data coding. An excellent comprehensive study of this is seen in Julian Ashwin, Aditya Chhabra and Vijayendra Rao’s (2025) article in *Sociological Research and Methods*. They conducted a multi-year survey and interview study with Rohingya refugees and their Bengali hosts in Bangladesh. They first analyzed the data with a detailed human-created coding scheme. Then working with several AI systems, they prompted the AI to analyze the data applying their coding scheme. What they found was “that LLMs over-predict codes that are very sparse (i.e. there are many more false positives than false negatives)” and “that in many cases LLM prediction errors are systematically associated with characteristics of the interview subject (e.g. refugee status, gender, education)” (np). As they explained:

If the codes generated by LLMs are not completely accurate, this is not necessarily a consequential problem. If the mistakes they make are random, with a large enough sample we should still be able to come to correct conclusions. However, if the mistakes are not random, then using LLM coding could lead to completely incorrect conclusions. In other words, if the LLM errors are biased, then relying on these codes could lead researchers to identify relationships in the data that are purely a result of algorithmic biases rather than reality. (Ashwin et al., 2025, np)

What Ashwin, Chhabra, and Rao recommend is that researchers seeking to leverage genAI for analysis use smaller, customized models that have been carefully trained on curated inputs.

But curated inputs may not fully solve the problem, particularly given some genAI models’ tendency toward sycophancy. For example, Miryam Naddaf (2025b), in a short but excellent article “AI chatbots are sycophants—researchers say it’s harming science”, asked researchers who use AI “how its propensity for people-pleasing affects their work—and what they’re doing to mitigate it.” Across

various AI systems, researchers reported that AI placed too much value on what the human entered in a prompt was accurate, leading to models that confirm what was input rather than identify the errors in the input (Petrov et al., 2025). As discussed in the article, Marinka Zitnik, a researcher in biomedical informatics at Harvard University, used multiple AI agents in research to address this problem, assigning “different roles to AI agents — for example, tasking one agent with proposing ideas and getting another to act as a sceptical scientist to challenge those ideas, spot errors and present contradictory evidence” (Naddaf, 2025b). Regardless of the strategies to lessen bias, the biases will always exist, and as researchers we have an ethical imperative in our human-applied analysis and in any use of AI systems to identify and mitigate bias as much as possible.

Dangers of homogenization & loss of critical thinking

LLMs can potentially flatten thinking, often privileging perspectives already privileged (Alvero et al., 2024). In addition, while genAI can be helpful for generating ideas, those ideas can be merely regurgitated and are not necessarily new, lessening creativity (Anderson et al., 2024) and giving researchers the illusion of understanding their data and the research questions being asked (Messeri & Crockett, 2024). This could lead to a narrowing and sameness of research foci and study design. In addition, if researchers overrely on AI for research, particularly in analysis, critical thinking and skill development could be impacted, as has been reported in other fields (Lee et al., 2025, see also Kosmyna et al., 2025).

This homogenization and loss could be particularly problematic for those who are newer to the research process, such as undergraduates or graduate students who do not have years of experience to draw from so as to leverage genAI as an assistant and being able to counterbalance AI suggestions. As a journal, *Computers and Composition* has, I believe, always sought to validate and support student agency in welcoming publications from graduate students and inviting graduate students to be guest editors of special issues. But as a field we now need, for experienced and new researchers alike, to create the spaces and processes to affirm human agency and decision-making in the research process. Certainly, one approach is to simply not use AI anywhere in the research and research writing process, and that is an option. Yet I think that might be too limiting and that we need to engage in critical exploration of AI, while also promoting critical reflection on what humans bring to the process. Just as some people were afraid of word-processing, worrying, for example, that the ease of cutting-and-pasting would lessen writers’ abilities to deeply engage with their texts, so too might it be that deeply engaging with genAI won’t lessen critical thinking but develop it in new directions. We need more research on the impacts of AI on research.

Environmental impacts

All digital technologies have serious environmental impacts. In terms of material design, the laptops and cell phones that we use depend on mining extraction and the exploitive human labor that goes with it. And it’s not just in the making of computer technologies, but in the recycling at the end of life too, a point made over a decade ago by Shawn and Kristi Apostel (2009). I mention this not to lessen the even more significant impacts of AI, but to note that it’s a problem we as a field have had to reckon with for a long time, and we’re going to face it even more in the future because of AI.

In terms of energy usage, data centers are proliferating across the country, impacting people, communities, and ecosystems in negative ways (Edwards, 2025a; Edwards, 2025b). In addition to noise pollution, because of the processing power needed for the massive number of servers, data centers use a ton of electricity. Different AI models have different levels of energy usage (Jegham et al., 2025) with some being a bit more efficient than others, but that changes with every new model released. Finding the energy usage of different systems is also challenging, as increasingly AI companies are burying that information. But it’s estimated (as of September 2025) that ChatGPT uses almost 40 million kwh per day, enough to charge 8 million cell phones; and, in a year, ChatGPT uses enough electricity to have powered all the homes in Iowa (Business Energy UK, 2025). The amount of power needed is leading to even more use of fossil fuels and the creation of more power plants, including the largest gas-powered power plant in the United States being built outside of Pittsburgh (Phillips, 2025). Because data centers need continuous high flow of electricity, the use of solar and wind is not always an option.

Water usage is even more disturbing, given what a finite resource access to clean, fresh drinking water is. Companies such as Google and Microsoft regularly publish the water usage for their data centers, but that information too is hard to find with some companies, like Amazon, being accused of intentionally misleading communities about the water usage their centers require (Barratt & Furneaux, 2025). It’s estimated that a large data center can use up to 5 million gallons of water a day (Yañez-Barnuevo, 2025). As researchers, as we consider whether to integrate genAI into our research processes, we need to consider the environmental impacts of such a decision and, if we decide to proceed, to seek out systems and companies that have better environmental records and transparency.

Given these challenges, why would any researcher use AI?

So far, I’ve discussed a lot in relation to threats and challenges posed by genAI and the technological infrastructure to support AI systems, and this raises the very real question: why would any researcher even consider using AI then?

I certainly do not advocate that all researchers must integrate AI into their research. There are important considerations for an ethic of refusal (e.g., Edwards, 2025b; Sano-Franchini et al., 2025). But as I reflect on the last 40 years of *Computers and Composition* research we have often charted a pathway negotiating critique, exploration, and application of new technologies, what one reviewer of this manuscript, described as a “middle way” an approach shaped by careful, critical exploration to see where and how we might support teaching, learning and researching with new digital technologies. As Andrew Feenberg articulated in *Critical Theory of Technology*, “technology is not a destiny but a scene of struggle. It is a social battlefield, or perhaps a better metaphor would be a parliament of things on which civilizational alternatives are debated and decided” (p. 14). As a field, we certainly must join this struggle and

civilizational discussion of AI technologies, especially because AI feels different in the challenges it presents, and it feels more threatening to individuals and societies as it is being promoted by companies with seemingly unlimited funds and often, it seems, in alignment with authoritarian approaches to governing with increased surveillance and automation (Cevallos, 2025; Garbe & Maerz, 2026).

I think that we as critical technology researchers (and educators who teach research) have an ethical responsibility to engage with new technologies, recognizing that we are shaped by and shaping of technologies and that our research can provide important and needed insights into genAI and how it might be used and designed differently and into how it might be integrated in research in ways that are beneficial. *Critique* has always been essential for our work with new technologies, but we also need *design*, as Kress (1999) noted in relation to a then new technology of multimodal composing. Applying that to genAI and research, we cannot assist researchers (be they students, academic colleagues, industry professionals) in the research and writing processes they are using—and many are using AI—without at least some researchers in the field of computers and writing working with AI and studying and applying it in research to see its impacts and how it may and may not be helpful. I think, as a field, that is a responsibility we have just as we have taken on the responsibility of studying previous emerging technologies such as word-processing, the Internet, the Web, social media, mobile technologies, and more.

I also think there are some “good consequences”—to return to Hugh Burns’ words from 1983—in the use of genAI to assist in some phases of the research process. But such usage requires careful consideration of the contexts and expectations for research and of the opportunities and challenges that arise from usage throughout the process. We must ask *cui bono*? as Kris Blair and Elizabeth Monske (2003) did in the 20th-anniversary special issue of *Computers and Composition*. They were examining the promises and perils of online learning, but their question applies to considerations of all technologies. Researchers in computers and writing have already begun extensive research on the impacts and uses of genAI for writing instruction, and we are now beginning to turn our attention to genAI usage for research and research processes.

Using AI to assist in the research process

Obviously not all uses of LLMs and generative AI are malevolent and dishonest uses. Many researchers are using AI in ways that align with ethical and responsible research practices, and I think there absolutely is such a thing as responsible use of AI for research, although I think there need to be guiding principles in place for that use. I will begin this section by discussing some possible guiding principles. Then I will turn to a brief discussion of integrating AI in various phases of the research process, providing examples drawn from and relevant to computers and writing research.

Transparency and disclosure

Transparency is the first such guiding principle. As educators, many of us advocate for students to be transparent about their use of AI in their scholarship. Student researchers (undergraduate and graduate) who are conducting and reporting on research as part of a course or degree program requirement need to consult closely with faculty members to ensure that any AI usage for research meets the course/program expectations and requirements. That same ethic of transparency for student use of AI applies to faculty and professional research as well. As researchers, we should disclose when and how we’ve used (or plan to use) genAI in the process of conducting or writing up our research. This isn’t just an ethic; in some cases, with some funding agencies and publication venues, it’s a requirement.

When writing proposals for funding, it’s imperative to be aware of the AI policies of the agency. Many private foundations, local and state organizations, and federal agencies require disclosing the use of AI to help write a submitted proposal, and they have restrictions on how AI may be used in the process. For example, the [National Endowment for the Humanities \(2024\)](#) has a policy on AI that “will require all applicants to specifically acknowledge (by use of footnotes or other marginal notations) when they have inserted AI-generated text into their proposals.”

Being transparent in AI usage also applies to publishing research. Many journals require an AI disclosure statement, as does *Computers and Composition*, whose requirement states: “Authors must declare the use of generative AI in the manuscript preparation process upon submission of the paper.” A more extensive generative AI policy statement and one that other journals may wish to emulate is from [College Composition and Communication \(2025\)](#), which focuses first on authorship and accountability, calling on authors to be transparent in usage and to “confirm that you are responsible for the originality and intellectual integrity of your submission, own the rights to any content produced by GenAI, and vouch for the ethicality and integrity of any statements or citations created by GenAI.” Like many journals, CCC also provides guidelines for reviewers, noting that reviewers must maintain manuscript confidentiality and that they may not upload any part of or quotations from a manuscript to genAI, but that reviewers may use genAI to assist them in writing their review. Unlike many journals, CCC also explicitly calls out AI refusal and potential bias against those who use AI: “Reviewers who object to engaging with work that involves GenAI, due to professional or ethical beliefs, are encouraged to decline the review invitation” so that editors “ensure that disclosure is not a basis for bias while making final decisions.” Given how strongly some people feel about any use of AI, this explicit request for transparency around refusal is important and aims, as the CCC statement explains, to “respect[] professional boundaries while maintaining fairness in review assignments.” Ultimately, the CCC statement asks that “all parties are accountable for upholding the values of human-centered scholarship and transparent engagement”—important principles for anyone involved in research and digital technologies.

And, finally, the ethics of transparency applies as well when communicating with research participants in human subjects research. Although there is not a regulation per se related to AI in 45CFR46, I would argue that the Belmont Principle of Autonomy which shapes

the regulation that participants need to be fully informed of what they are consenting to applies to the use of AI in research. In consent forms, researchers should inform potential participants that their data may be entered into genAI systems and what the privacy settings are for those systems. For example, adding a statement such as: “De-identified data from the [survey, interviews, observation notes, coursework, etc.] may be entered into [name of AI system/s] for analysis. The account(s) used for this have been configured to not use data uploaded for AI training purposes.” The latter point about privacy settings I’ll discuss in a moment but next are two important areas of guidelines to consider.

Agency and autonomy

Key to any AI usage in any endeavor is centering human agency and ensuring that humans are making the decisions in the research process. Researchers need to always center their own human agency and to recognize AI as an assistive tool, not as the researcher. To return to *Computers and Composition*’s required disclosure statement as a guide: it states:

Authors preparing a manuscript for an Elsevier journal can use AI Tools to support them. However, these tools must never be used as a substitute for human critical thinking, expertise and evaluation. AI technology should always be applied with human oversight and control.

Ultimately, authors are responsible and accountable for the contents of their work. [...] Finally, authors must not list or cite AI Tools as an author or co-author on the manuscript since authorship implies responsibilities and tasks that can only be attributed to, and performed by, humans (Elsevier, n.d).

Emphasized across most AI policies and requirements is the ethic of human agency and that the human researcher or research team is always responsible for the research conducted and the results disseminated.

Not discussed as much, but an area that in the future is going to become even more pressing is autonomy. When using genAI systems, researchers need to consider what level of autonomy and agency they will allow the machine and for what tasks. When can something in the research process be machine-only, when human-only, and when various combinations of both (see Simmler & Frischknecht, 2021)? As we now move into agentic AI, where AI systems on their own leverage different platforms and programs to complete tasks, we will increasingly be faced with decisions around, “Do I trust this AI system to have access to my computer programs to complete this task?” Ethan Mollick (2026) offers a useful overview of the components at work with agentic AI: *models*, the underlying systems of the LLM; *apps*, the platforms/places where these systems work—most of us are used to web-based or phone-based chat platforms where we dialogue with LLMs; and *harnesses*, the connective programming that allow genAI to work across systems, like how ChatGPT can search the web or create Microsoft Office documents. I do not have time in this article to discuss more fully the implications of agentic AI—and the implications are evolving quickly as the models get more and more robust and are able to work across and within more systems—but I will just mention that the move to agentic means we as researchers will need to consider more than ever when, where, and how should machines be used in the research process.

Privacy, data security, & intellectual property

LLMs have varying approaches to how text and other information inputted are treated potentially exposing research data to harms (Chen et al., 2025), including harms to research participants, to the intellectual property of the researcher, and, as applicable, to the original creators of the research data. Researchers need to recognize that AI systems exist in and are integrated into platforms in a variety of ways. Some AI systems a researcher may use are embedded into pre-existing platforms (e.g., Qualtrics, Nvivo) that have policies that have long been vetted for research and intellectual property protections. Other systems are hosted locally by a university on the university servers and are thus controlled by the researcher (customized Small Language Models, for example). But some systems exist in cloud-based hosted systems where the data is going to a third-party server (e.g., using web-based OpenAI ChatGPT, Anthropic Claude, Google Gemini, etc.). Many genAI systems do not have privacy settings or terms of service that align with research ethics, particularly in regard to human subjects research (with the need to follow 45CFR46), student research (with FERPA requirements, 39CFR99), and intellectual property considerations. For many AI systems, the entries provided by users become part of the systems’ training data and may become available in some form to other users thus exposing a researcher’s data and raising potential violations of privacy, confidentiality, and intellectual property.

Thus, as researchers, we have an ethical responsibility to know the terms of service and privacy policies for any AI systems we might use in research (including transcription systems), and we all have a responsibility to keep updated on the system’s ever-changing policies and settings. For example, Anthropic (2025) changed their policy so that paid account users’ inputs now go to the systems training data unless a user opt-outs. Because of the challenge of ever-changing policies that AI companies can legally change at will, researchers should consider using enterprise-level accounts that their institution or organization might have contracted for and thus have terms set by a legally binding contract that is not changeable at will. For educational institutions, the enterprise-level systems, like Google’s Gemini or Microsoft’s Copilot, come with higher educational protections that ensure no information entered go into more widely available training sets. However, given the proliferation of AI tools and the diverse needs we face in research, we may find that if we wish to use AI, we need to use a non-enterprise-level system. When that’s the case, asking and answering questions such as the following can be helpful:

- How protected is your data (and your subjects' data)?
- Who (government officials, third-party vendors, etc.) may access your data and for what reasons?
- Will your data be used to further train the systems?
- Will you be notified if the terms of service/privacy policies change, and, if they do, can you pull and remove all data?
- What are your rights as a user of the system?
- And, importantly, do you own the intellectual property of any outputs of the system?

These are questions we've asked of other digital technologies, including social media, and, as we know from our teaching, involves best practices to ensure informed, critical use of technological platforms.

Possible integrations of AI in the research process

There are a number of ways researchers might wish to use AI in the research process but it's crucial to remember that each of these uses occurs within the very real and problematic limitations of AI and the flaws of LLM design models. Human researchers need to verify and confirm the validity of any AI outputs and guard carefully against biases and misrepresentations. In addition, as discussed, researchers must consider what harms may arise to the privacy, confidentiality, and intellectual property of their research. Sometimes the potential benefits of the speed and computing power of genAI are not worth the risks, and I would urge any researcher not to share research data or research write-ups with AI systems where inputs become available for model training (and thus model output to other users) and where the AI company may claim intellectual property rights over any material inputted or outputted from the system.

The opportunities to design LLMs for research are extensive and connect with a long tradition in computers and composition (going back to the 1970s and 1980s with TOPOI and Daedalus) where scholars have contributed to digital system design and development. When working with commercial or enterprise AI systems researchers may tailor LLMs by creating, for example, Google Gems, Custom GPTs, Claude Projects, that are trained on specific material and work within bounded parameters for analysis and output. For researchers with more development skills and access to computing resources there is also the opportunity to build new models rather than work with existing base models. As computers and writing scholars, we will benefit from seeking interdisciplinary collaborations with colleagues in computer science and other programs to explore customization further. I mention design because in the list of possible uses in Table 1, any of these processes could be further customized and applied using a tailored LLM or a new model designed specifically for the type of research being conducted.

Table 1

Possible usage of genAI to assist in research processes.

<i>Researchers should not put their research data or their intellectual property into an AI system without ensuring that the privacy settings and terms of service are set to strict confidentiality and protection of data.</i> Planning/Proposal • Locate, analyze, and summarize secondary research • Assist with literature reviews • Identify possible sources of funding and prepare funding proposals (within the guidelines and requirements of the agency) • Complete IRB or other regulatory approval applications • Consider possible methods and methodologies to apply and ethical challenges that may arise (with, of course, one of those challenges being in the very use of the system if it leads to homogenization of research) • Identify possible sites, populations, data sources of study • Design survey/interview questions and other data collection protocols • Provide feedback and ideas as an algorithmic reviewer Data Collection • Create customized models for specific tasks • Locate and recruit participants for research (with caveat that this only works in some contexts and has the potential for far more problems) • Create survey follow-up questions on the fly, customized based on the real-time response of each respondent (potential problems but an area to keep researching) • Provide synchronous or asynchronous translation • Create automatic transcripts of interviews and other data recordings • Conduct video/phone interviews (particularly structured, but, again, a caveat that this may not garner results aimed for) • Create synthetic data or serving as synthetic subjects for some types of research (recognizing the ethical challenges and biases of synthesis) Data Analysis • Clean data sets and code data • Create data codes or providing feedback/revisions to existing codes • Identify outliers, areas for further analysis etc. • Connect and compare data sets • Provide insights and findings from data analysis Data Reporting • Identify venues for publication or conference presentation • Contribute to all phases of the writing process—from outlining and drafting to revising and copy-editing • Offer feedback on study reporting based on evaluation criteria used by the publication venue (e.g., journal, funding agency, fellowship competition etc.) • Create data visuals and presentation slides
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A few examples of usage in action

At the start of a project when reviewing literature, genAI systems can help researchers locate studies and scholarship in ways more integrative than simple Google Scholar or library database searches. I have leveraged genAI by asking versions of, “I am interested in studying X exploring Y with Z populations. What studies have been conducted that may relate to this?” I have found that genAI systems provide scholarship in fields or from regions in the world that I have not as often searched for. Now, it is true, many sources identified by genAI are either commercial hype or not as strong scholarly sources, but amid the not-so-helpful returns are some sources that I know I would not have otherwise found.

Dialoging with genAI throughout the research process can be helpful to explore possibilities for research and to prompt reflection on one’s own research processes. As we all know, reflection is a key part of research, and a study published in the *Journal of Technical Writing and Communication* (Thominet et al., 2024) found that conversing with genAI in various roles (manager, teacher, colleague, advocate) and at various points of the process can assist qualitative researchers in reflecting on their approaches to research and to working with AI in the research process. This makes me think of the years I spent involved with the Qualitative Research Network (QRN), an annual forum for researchers to get feedback from other researchers on their work. Many of the researchers who participated in the QRN were at smaller schools, sometimes where they were the only rhetoric and writing faculty member. Not all of us are immersed in extensive research communities with colleagues down-the-hall to be able to discuss our research with and because of the privacy and confidentiality of a lot of research we may not as easily be able to post or discuss online. Having a genAI system (with privacy settings and protections) to dialogue and consult with is not a substitute for dialoguing with human researchers but it is another assistive avenue for research feedback and reflection.

In relation to data collection, some researchers are experimenting with AI interviewers so as to more easily scale projects, and systems such as Qualtrics offer survey customization powered by AI so that follow up questions are generated in real-time based on respondents’ answers. Call me traditional, but giving too much agency to AI systems for data collection in person-based studies makes me uneasy. There is an art to interviewing that I do not think genAI (at least in its current stage of development) would be able to master. My feeling of uneasiness is supported by some research. Researchers from Microsoft and several universities (Cuevas et al., 2025) evaluated three different chatbots for interviewing people ($n = 399$), measuring participant engagement and the “richness” of the responses in terms of complexity and specificity of social context. They found that “while the chatbots were able to elicit high-quality responses based on established evaluation metrics, the responses rarely capture participants’ specific motives or personalized examples, and thus perform poorly with respect to richness.” They also found lower interrater reliability between LLMs and humans, and they conclude that “Our study offers a cautionary tale for scaling and evaluating qualitative research with LLMs.”

I don’t believe many researchers in writing studies have tried genAI for interviewing, but many of us who conduct interviews have probably used genAI for transcription. In my uses of AI for transcription, I’ve found there have been gains and losses. For much of my career because I did not have research funds for paid transcription, I self-transcribed recorded interviews. I’d spend hours scrubbing through files, able to transcribe a 45-minute interview in about 2–3 h. On the one hand that was a lot of time, but on the other, it was time well spent. I was deeply immersed in the interview, listening again and again to participants’ voices and stories so that I knew so well what was said. I could pull up from memory participants’ stories to integrate into manuscripts on first drafting with just writing and then going back later to verify and correct. But now, for the past interview project I did, I used Zoom’s built in AI transcription service (from Miami’s enterprise contract and with participants’ consent). After the interview, I would relisten to the audio (also recorded) and verify the transcript, but that was it. I did not dwell with the interviews as deeply or for as long. So, I found that I then needed to take the time to read and re-read and read again the transcripts. In the end I think I gained the same level of depth, but it was a different process and, while I do not want to return to the long hours of transcription, I do want to be sure that I don’t lose extended engagement with interview participants via reading and re-reading transcripts.

GenAI may assist in coding qualitative data, as is being explored by researchers in many fields, and as I discussed earlier in relation to bias with the study by Ashwin, Chhabra and Rao (2025), where they compared human coding and AI-coding of extensive survey and interview data from Rohingya refugees. In another example, Mohammad Jalali and Ali Akhavan (2024) conducted a replication study where they returned to an earlier study and had an LLM (ChatGPT 4) recode the data to compare their original human coding to the LLM coding. What they found was that the AI coding lacked some key contextual insights but that it also provided some approaches for analysis the researchers had not originally considered. They concluded that genAI has “great potential to serve as an augmentative tool rather than a replacement for the intricate analytical tasks performed by humans.” What I find particularly compelling about both of these studies in terms of method is the replication and human-machine comparison.

I have only tried AI for data coding once so far with a Miami University Institutional research project. Working with the Provost’s office, colleagues and I created a survey that was sent to every student at Miami University. In our IRB-reviewed consent form, we told participants that the text-entry responses to questions would be entered into Google Gemini (the system Miami has an enterprise-level contract with) as another form of analysis. When I entered the qualitative data into Google Gemini, I found that Gemini was able to identify a number of themes that I as a human coder had also identified, but that it also missed consideration of some responses, particularly outlier responses or briefer responses, often written in less detail and with more grammatical errors, that showed important perspectives. Now, some of that could have been my prompting, that I didn’t specify enough to look for outliers, but some of that is too that it seemed that Gemini privileged the longer responses where students had written a lot more and in more “standard” English. Going forward, I think computers and composition research would benefit from further comparison and replication studies looking at human coding and AI coding of data, as scholars like Ashwin, Chhabra and Rao (2025) and Jalali and Akhavan (2024) are doing, so as to identify the biases and weaknesses in the various AI systems when working on different coding tasks and also to identify where, if any, strengths lie.

GenAI can, of course, be helpful in writing up research. Cheng et al. (2025) examined how ESL graduate students leveraged ChatGPT to assist with research writing. As they explained, their study “highlights how advanced ESL writers can adapt GenAI tools to their unique writing processes” and their findings “suggest that AI integration in academic writing can be customized to support diverse writing goals and backgrounds.” Just as there is no *one* writing process, as years of studies on writers and their processes have shown, so too is there no *one* way to approach integrating AI in research: approaches will differ by researcher, by research question, research site, population, and context, and by research methods and methodologies.

I'll close this section with one other important area for us to consider: overreliance (Gallagher, Wagner & Canzonetta, 2025; McKee, 2026). When do researchers find that they are overrelying on AI, possibly threatening the integrity of their research analysis or research reporting? How much AI usage is too much AI usage? What level of usage in what contexts will we as a field find acceptable? And for some researchers the short and firm answer will be *any* usage is too much because for some researchers the environmental threats, for example, are just too great. But for other researchers, and I put myself in this camp, the pragmatic recognition is that AI use is here and being integrated into every field, and we gain by studying what these systems can and cannot do in terms of helping in the research process. We might find ways to approach research that we could not do or could not do as easily before with AI. There is much we all have to learn and explore, especially as AI opens new research directions.

New research directions

In terms of research, there are certainly all the research questions about the usage, ethics, and impacts of AI on writing and how writers are integrating (and resisting) genAI across contexts. But keeping the focus on research, I want to consider what new research directions *about research* we might consider as a field in this age of generative AI.

As I mentioned above, we need more studies in every phase of the research process examining the impacts, challenges and opportunities of AI integration. What is gained, and what is lost when AI is used for, say, literature reviews, for drafting grant applications, for designing surveys, for coding data, and for writing reports? We need so many more studies and reflections on researchers' experiences working with AI. We also need replication studies so we can compare human-only and human-machine and even, possible, machine-only research processes. What are humans good at in research? What is genAI good at? How might we work with AI to further research?

We also need to consider the teaching of research and issues of deskilling (see Lee et al., 2025) and skills learning. I teach research methods to undergraduate and graduate students, and I struggle with where and when to introduce possible integration of AI in the process. What areas of research methods should always be human-only or at least human-only on first learning? How can we research educators ensure that new researchers learn the skills they need to then be able to work critically and with their own experience when working with AI if they leverage AI from the get-go? If you've never yourself coded data and genAI produces codes for you in seconds, how do you gain the expertise and perspective to see the problems, gaps, biases in what might be produced? My answers now to these pressing questions are very much *it depends*, upon the researcher, the research context, the research questions. But what I know is a constant is that researchers must always critically and reflexively analyze their research positionality and their approaches to research—whether genAI is involved in the process or not. We need more studies of how we teach research methods in an age of AI.

And we need studies of research processes in multiple contexts, including, importantly, professional contexts. Amber Hedquist, Mark Hannah and Heidi Willers (2026) conducted a study on the emerging role of technical communicators as AI facilitators in research workflow, studying 422 dialogical interactions between a technical communicator and an AI system. What they found was that “AI Facilitators enable the AI to participate more meaningfully in the co-construction of knowledge. These pivots reflect the adaptive, rhetorical work required to sustain collaborative engagement, highlighting a growing opportunity for TCers [technical communicators] to apply and expand their expertise in AI-enabled spaces” (p. 47). New roles in research will open up with the integration of AI. Our field, because of our long history of critical analysis and rhetorical adaptability, is well positioned to help prepare researchers for these openings. We are also well positioned to study the design of AI systems, including looking for opportunities to study the design of customized approaches within existing models and the development of new models. How these systems are created and who is involved in the design decisions matters greatly, as work by many early developers of AI have noted (see Buolamwini, 2023; Bender et al., 2021). As we as individuals and as a field learn more and as we create more customized approaches to AI and research, we will certainly have valuable insights and approaches to share, particularly if such approaches can lessen the potential harms and challenges of genAI.

Creating and shaping future computers and composition research

Cynthia Selfe and Kate Kiefer in 1983 launched what became the journal *Computers and Composition* with a vision for what was needed to help our then newly emerging field grow and explore questions of import. So many of us have gained so much from *Computers and Composition* over the years, and the studies published in the journal have helped us build and share knowledge.

If we wish to sustain the field and the journal over the next 40 years we need to continue to turn our attention to the impact of emerging technologies such as genAI, particularly addressing the threats AI poses to research processes and publication and considering the “good and bad consequences,” as Hugh Burns (1983) said, of AI usage by researchers. I share current *Computers and Composition* editor Jason Tham's optimism when he wrote in his September 2025 introduction: “Despite existing challenges, I approach the future with hopefulness and optimism. Technology holds the potential to foster deeper connections and more equitable educational environments. It is this hopeful vision that drives our collective work at C&C, encouraging us to continuously reflect, adapt, and innovate with computers and writing research.” (Tham, 2025).

I hope this article serves as invitation to researchers across all levels of experience and working in all contexts to critically reflect on, research, and, as appropriate, integrate AI in research. Our field needs more studies of AI, of its impacts and uses in research, of its design and potential, and of its harms. I look forward in the years to come to see what arises. Knowing our field, the work will be sharp, critical, and deeply engaged with ethical considerations, centering human agency and decision-making every step of the way.

AI disclosure statement

I used ChatGPT (5.2 Instant & Thinking) to assist with formatting references and to search for sources that I then read and considered.

CRedit authorship contribution statement

Heidi A. McKee: Writing – original draft.

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No data was used for the research described in the article.

References

- Adam, David. (2025). When AI rejects your grant proposal: Algorithms are helping to make funding decisions. *Nature*. <https://doi.org/10.1038/d41586-025-02852-9>
- Alvero, A. J., Lee, Jinsook, Regla-Vargas, Alejandra, Kizilcec, René F., Joachims, Thorsten, & Antonio, Anthony Lising (2024). Large language models, social demography, and hegemony: Comparing authorship in human and synthetic text. *Journal of Big Data*, 11(1), 138. <https://doi.org/10.1186/s40537-024-00986-7>
- Anderson, Barrett R., Shah, Jash H., & Kreminski, Max (2024). Homogenization effects of large language models on human creative ideation. In *Proceedings of the 16th Creativity & Cognition Conference (C&C '24)* (pp. 413–425). Association for Computing Machinery. <https://doi.org/10.1145/3635636.3656204>.
- Anthropic. (2025, August 15). Usage policy update. <https://www.anthropic.com/news/usage-policy-update>, <https://www.swiss-ai.org/apertus>.
- Apostel, Shawn, & Apostel, Kristi. (2009). Old world successes and new world challenges: Reducing the computer waste stream in the United States. *Technological ecologies & sustainability*. Computers & Composition Digital Press/Utah State University Press. https://ccdipress.org/tes/16_apostel_apostel.pdf.
- Ashwin, Julian, Chhabra, Aditya, & Rao, Vijayendra (2025). Using large language models for qualitative analysis can introduce serious bias. *Sociological Methods & Research*. <https://doi.org/10.1177/00491241251338246>. Advance online publication.
- Barratt, Luke, & Furneaux, Rosa. (2025). Amazon strategised about keeping its datacentres' full water use secret, leaked document shows. *The Guardian*. <https://www.theguardian.com/technology/2025/oct/25/amazon-datacentres-water-use-disclosure>.
- Bender, Emily M., Gebru, Timnit, McMillan Major, Angelina, & Shmitchell, Shmargaret (2021). On the dangers of stochastic parrots: Can language models be too big?. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)* (pp. 610–623). Association for Computing Machinery. <https://doi.org/10.1145/3442188.3445922>.
- Blair, Kristine L., & Monske, Elizabeth A. (2003). Cui bono?: Revisiting the promises and perils of online learning. *Computers and Composition*, 20(4), 441–453. <https://doi.org/10.1016/j.compcom.2003.08.016>
- Boboris, Kat. (2025). Attention authors: Updated practice for review articles and position papers in arXiv CS category. *arXiv Blog*. <https://blog.arxiv.org/2025/10/31/attention-authors-updated-practice-for-review-articles-and-position-papers-in-arxiv-cs-category/>.
- Buolamwini, Joy. (2023). *Unmasking AI: My mission to protect what is human in a world of machines*.
- Burns, Hugh L. (1979). *Stimulating rhetorical invention in English composition through computer-assisted instruction (dissertation)*. University of Texas at Austin. Available through the Internet Archive <https://archive.org/details/DTIC.ADA106372>.
- Burns, Hugh. (1983). A note on composition and artificial intelligence. *Computers and Composition*, 1(1), 3–4.
- Business Energy UK. (2025). *ChatGPT energy consumption visualised* [Month unknown]. Business Energy UK <https://www.businessenergyuk.com/knowledge-hub/chatgpt-energy-consumption-visualized/>.
- Cevallos, Albert. (2025). How autocrats weaponize AI — And how to fight back. *Journal of Democracy*. <https://www.journalofdemocracy.org/online-exclusive/how-autocrats-weaponize-ai-and-how-to-fight-back/>.
- Chen, Kang, Zhou, Xiuzhe, Lin, Yuanguo, Feng, Shibo, Shen, Li, & Wu, Pengcheng. (2025). A survey on privacy risks and protection in large language models. *arXiv*. <https://doi.org/10.48550/arXiv.2505.01976>.
- Cheng, Dongmei, Li, Mimi, & Lee, Tony. (2025). Leveraging ChatGPT for research writing: An exploration of ESL graduate students' practices. *Computers and Composition*, 76, Article 102934. <https://doi.org/10.1016/j.compcom.2025.102934>
- College Composition and Communication. (2025, October). CCCC generative AI policy statement. <https://cccc.ncte.org/cccc/generative-ai-policy>.
- Cuevas, Alejandro, Scurrell, Jennifer, V., Brown, Eva, M., Entenmann, Jason, ... Madeleine. (2025). Collecting qualitative data at scale with large language models: A case study. *Proceedings of the ACM on Human-Computer Interaction*, 9(2), CSCW049. <https://doi.org/10.1145/3710947>
- Dall, Chris. (2025). MAHA report on chronic disease in US kids includes fake citations and other errors. *CIDRAP*. <https://www.cidrap.umn.edu/public-health/maha-report-chronic-disease-us-kids-includes-fake-citations-other-errors>.
- Edwards, Dustin. (2025). *Enduring digital damage: Rhetorical reckonings for planetary survival*. University of Alabama Press. <https://www.uapress.ua.edu/9780817322472/enduring-digital-damage/>.
- Elliott, Oli. (2025). *Representation of bbc news content in ai assistants*. BBC. <https://www.bbc.co.uk/aboutthebbc/documents/bbc-research-into-ai-assistants.pdf>.
- Elsevier. (n.d.). Guide for authors, computers & composition. Retrieved from <https://www.sciencedirect.com/journal/computers-and-composition/publish/guide-for-authors>.
- Ericsson, Patricia Freitag, & Haswell, Richard H. (Eds.). (2006). *Machine scoring of student essays: Truth and consequences*. Logan, UT: Utah State University Press.
- Fernandes, Maggie, & McIntyre, Megan. (2025). Drafting defensively, documenting authorship: An analysis of draftback and Grammarly authorship. *Computers and Composition*, 76. <https://doi.org/10.1016/j.compcom.2025.102926>
- Gallagher, John R., Wagner, Kyle, & Canzonetta, Jordan. (2025). When collaborating turns into dishonesty: A data-driven heuristic comparing human and AI collaborators. *Computers and Composition*, 77. <https://doi.org/10.1016/j.compcom.2025.102947>
- Garbe, Lisa, & Maerz, Seraphine F (2026). The rise of authoritarian informationalism: Escalating surveillance, manipulation, and control. *Democratization*, 33(1), 1–18. <https://doi.org/10.1080/13510347.2025.2579105>

- Haider, Jutta, Söderström, Kristofer Rolf, Ekström, Björn, & Rödl, Malte (2024). GPT-fabricated scientific papers on Google Scholar: Key features, spread, and implications for preempting evidence manipulation. *Misinformation Review*, 5(5). <https://doi.org/10.37016/mr-2020-156>
- Hedquist, Amber, Hannah, Mark A., & Willers, Heidi. (2026). Navigating human-AI collaboration: The emerging role of technical communicators as AI facilitators. In Miriam F. Williams, & Lisa Melonçon (Eds.), *Liminality: The work of resilience in technical communication* (pp. 31–49). The WAC Clearinghouse/University Press of Colorado. <https://doi.org/10.37514/TPC-B.2026.2869.2.02>.
- Helgeson, Scott A., Johnson, Patrick W., Gopikrishnan, Nilaa, Koirala, Tapendra, Moreno-Franco, Pablo, Carter, Rickey E., Quicksall, Zachary S., & Burger, Charles D (2025). Human reviewers' ability to differentiate human-authored vs artificial intelligence-generated medical manuscripts: A randomized survey study. *Mayo Clinic Proceedings*, 100(4), 622–633. <https://doi.org/10.1016/j.mayocp.2024.08.029>
- Hofmann, Valentin, Kalluri, Pratyusha R., Jurafsky, Dan, & King, Sharese (2024). AI generates covertly racist decisions about people based on their dialect. *Nature*, 633, 147–154. <https://doi.org/10.1038/s41586-024-07856-5>
- National Endowment for the Humanities. (2024). Policy on the use of artificial intelligence for NEH grant proposals. National Endowment For The Humanities. https://www.neh.gov/sites/default/files/2024-10/NEH.AI_Policy-10.23.24.pdf.
- Jalali, Mohammad S., & Akhavan, Ali. (2024). Integrating AI language models in qualitative research: Replicating interview data analysis with ChatGPT. *System Dynamics Review*, 40(3). <https://doi.org/10.1002/sdr.1772>
- Jegham, Nidhal, Abdelatti, Marwan, Elmoubarki, Lassad, & Hendawi, Abdeltawab. (2025). How hungry is AI? Benchmarking energy, water, and carbon footprint of LLM inference. arXiv. <https://doi.org/10.48550/arXiv.2505.09598>.
- Kalai, Adam Tauman, Nachum, Ofir, Vempala, Santosh S., & Zhang, Edwin. (2025). Why language models hallucinate arXiv. <https://doi.org/10.48550/arXiv.2509.04664>.
- Kosmyna, Nataliya, Hauptmann, Eugene, Yuan, Ye Tong, Situ, Jessica, Liao, Xian-Hao, Beresnitzky, Ashly Vivian, Braunstein, Iris, & Maes, Pattie. (2025). Your brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing task. arXiv. <https://doi.org/10.48550/arXiv.2506.08872>.
- Kotwal, Ketan, & Marcel, Sébastien. (2025). Review of demographic fairness in face recognition. arXiv. <https://doi.org/10.48550/arXiv.2502.02309>.
- Kress, Gunther. (1999). Multimodality. In Gail E. Hawisher & Cynthia L. Selfe (Eds.), *Passions, pedagogies, and 21st century technologies*. Utah State University Press. <https://wacclearinghouse.org/books/usu/passions/>.
- Lee, Hao-Ping, (Hank), Sarkar, Advait, Tankelevitch, Lev, Drosos, Ian, Rintel, Sean, Banks, Richard, & Wilson, Nicholas (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence—Effects from a survey of knowledge workers. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)* (pp. 1–22). Association for Computing Machinery. <https://doi.org/10.1145/3706598.3713778>. Article 1121.
- Májovský, Martin, Černý, Martin, Kasal, Matěj, Komarc, Martin, & Netuka, David (2023). Artificial intelligence can generate fraudulent but authentic-looking scientific medical articles: Pandora's box has been opened. *Journal of Medical Internet Research*, 25, Article e46924. <https://doi.org/10.2196/46924>
- Maupin, Danny, Suchak, Tulsī, Barnett, Adrian, & Spick, Matt (2025). Dramatic increases in redundant publications in the generative AI era. *medRxiv*. <https://doi.org/10.1101/2025.09.09.25335401>
- McIntyre, Megan. (2025). Equitable writing classrooms and programs in the shadow of AI. *Computers and Composition*, 75. <https://doi.org/10.1016/j.compcom.2024.102908>
- McKee, Heidi A., & DeVoss, Danielle Nicole (Eds.). (2007). *Digital writing research: Technologies, methodologies, and ethical issues*. Hampton Press.
- McKee, Heidi A. (2026). Generative AI and college writing: Critical literacies and student perspectives on usage, course policies, and authorial balance. *Journal on Excellence in College Teaching*, 37(SpecialIssue), 43–58.
- Messeri, Lisa, & Crockett, M. J. (2024). Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627, 49–58. <https://doi.org/10.1038/s41586-024-07146-0>
- Mollick, Ethan. (2026, February 18). A guide to which AI to use in the agentic era. One Useful Thing. <https://www.oneusefulting.org/p/a-guide-to-which-ai-to-use-in-the-agentic-era/>.
- Naddaf, Miryam. (2025a). Journals infiltrated with 'copycat' papers that can be written by AI. *Nature*. <https://doi.org/10.1038/d41586-025-03046-z>
- Naddaf, Miryam. (2025b). AI chatbots are sycophants — Researchers say it's harming science. *Nature*. <https://doi.org/10.1038/d41586-025-03390-0>
- Petrov, Ivo, Dekoninck, Jasper, & Vechev, Martin (2025). BrokenMath: A benchmark for sycophancy in theorem proving with LLMs. arXiv. <https://doi.org/10.48550/arXiv.2510.04721>
- Phillips, Ari. (2025). America's largest gas plant grows out of artificial intelligence's hunger for power. *Oil & Gas Watch*. <https://news.oilandgaswatch.org/post/driven-by-ais-growing-hunger-for-electricity-americas-largest-gas-plant-is-planned-for-pa>.
- Prada, Luis. (2025). The CDC got caught citing a fake study. *Again*. VICE. <https://www.vice.com/en/article/the-cdc-got-caught-citing-a-fake-study-again/>.
- Sano-Franchini, Jennifer, McIntyre, Megan, & Fernandes, Maggie (2025). Recognizing AI hype masquerading as research. *Refusing GenAI in Writing Studies*. <https://refusal.blog/2025/05/21/recognizing-ai-hype-masquerading-as-research/>.
- Selfe, Cynthia L., & Kiefer, Kathleen. (1983). Letter from the editors. *Computers and Composition*, 1(1), 1.
- Simmler, Monika, & Frischknecht, Ruth. (2021). A taxonomy of human-machine collaboration: Capturing automation and technical autonomy. *AI & Society*, 36, 239–250. <https://doi.org/10.1007/s00146-020-01004-z>
- Spinellis, Diomidis. (2025). False authorship: An explorative case study around an AI-generated article published under my name. *Research Integrity and Peer Review*, 10. <https://doi.org/10.1186/s41073-025-00165-z>. Article 8.
- Sullivan, Patricia, & Porter, James E. (1997). *Opening spaces: Writing technologies and critical research practices*. Ablex.
- Tham, Jason. (2025). Editorial: Technology's double edge. *Computers and Composition*, 77.
- Thominet, Luke, Amorim, Jacqueline, Acosta, Kristine, & Sohan, Vanessa K (2024). Role play: Conversational roles as a framework for reflexive practice in AI-assisted qualitative research. *Journal of Technical Writing and Communication*, 54(4), 396–418. <https://doi.org/10.1177/00472816241260044>
- Xu, Ziwei, Jain, Sanjay, & Kankanhalli, Mohan (2025). Hallucination is inevitable: An innate limitation of large language models. arXiv. <https://doi.org/10.48550/arXiv.2401.11817>
- Yañez-Barnuevo, Miguel. (2025). Data centers and water consumption. *Environmental and Energy Study Institute*. <https://www.eesi.org/articles/view/data-centers-and-water-consumption>.